CRL Technical Symposium 2026

GYANOTSAV 2026 · Programme Schedule · BAE (Nalanda) · 11 & 12 November 2026

DAY 1 Wednesday, 11 November 2026

8:30–09:00	**Registration** Authors and Delegates
09:00–10:00	**Symposium Inauguration** Sir M Visvesvaraya Auditorium Address by Directors, CMD & Chief Guest
10:00–11.00	**Key Note Address** **Dr. Giridhar K, IIT Madras on "A Perspective on Proactive D&D for Wireless Systems"** Sir M Visvesvaraya Auditorium
11:00–11:30	**High Tea + Poster Presentation Visit**
11:30–13:15	**Technical Sessions 1–4** **Hall A** Session 1 · Computer Vision **Hall B** Session 2 · Microwave Components **Hall C** Session 3 · Signal Processing **Hall D** Session 4 · Detection & Classification
13:15–14:00	**Lunch Break**
14:00–15:00	**Invited Talk** **Shri Raghavendra M R, Director, ISTRAC on "Satellite Tracking and Command Network"** Sir M Visvesvaraya Auditorium
15:00–17:15	**Technical Sessions 5–8** **Hall A** Session 5 · Machine Learning Applications **Hall B** Session 6 · RF Subsystems **Hall C** Session 7 · Product Design and Simulations **Hall D** Session 8 · Electronic Warfare **Tea Break + Poster Presentation Visit 15:30–16:00**

DAY 2 Thursday, 12 November 2026

08:30–09:00	**Registration** Authors and Delegates
09:00–10:00	**Invited Talk** **Shri Shesha Raghunathan, Strategy Lead, IBM on "Towards Quantum Advantage and Beyond"** Sir M Visvesvaraya Auditorium
10:00 -10:30	**Tea Break + Poster Presentation Visit**
10:30–12:45	**Technical Sessions 9–12** **Hall A** Session 9 · AI Powered Systems **Hall B** Session 10 · Crypto & Cyber Security **Hall C** Session 11 · Communication Systems **Hall D** Session 12 · Radar Systems
12:45–14:00	**Lunch**
14:00–15:00	**Panel Discussion** **Topic: Next Generation Warfare - Ensuring India's Preparedness** **Moderator: Shri. Giriraja N, ED (EW&A)** **Panel: Smt. Dharani B CTO (R&WS), Shri. S K Mahajan GM (NS-R&FCS), Dr. Vishwanadham GM (EW-LS Hyd), Shri. Srinivasa Rao HP GM (IM), Shri. A K Singh GM (SP)**
15:00–15:15	**Tea Break + Poster Presentation Visit**
15:15–17:30	**Technical Sessions 13–16** **Hall A** Session 13 · Cognitive Systems **Hall B** Session 14 · Smart Computing Systems **Hall C** Session 15 · Autonomous Systems **Hall D** Session 16 · Photonics Systems
17:30-18:00	**High-Tea**

Venues

- **Hall A:** Dr. APJ Abdul Kalam Hall
- **Hall B:** Sir C. V. Raman Hall
- **Hall C:** Prof. Satish Dhawan Hall
- **Hall D:** Sir M Visvesvaraya Auditorium

DAY 1 · Wednesday, 11 November 2026 Technical Sessions 1–4 · 11:30–13:15 · Four Parallel Halls

TIME	HALL A · Dr. APJ Abdul Kalam Hall **Session 1: Computer Vision** Chair: Smt. Deepika Gupta, GM, DCCS-GAD Co-chair: Shri. Nandakumar S, AGM-NS	HALL B · Sir C. V. Raman Hall **Session 2: Microwave Components** Chair: Shri. Vrushabhendra S H M, In-GM, ANTENNA-GAD Co-chair: Shri. Prakash S P, AGM-PDIC	HALL C · Prof. Satish Dhawan Hall **Session 3: Signal Processing** Chair: Shri. Mini Nair S, GM-NS1 Co-chair: Shri. Jeyagopal P AGM Mkt MR	HALL D · Sir M Visvesvaraya Auditorium **Session 4: Detection & Classification** Chair: Smt. Seema Yadav, CS, CRL-GAD Co-chair: Shri. M Sadasivaiah AGM EW&A
11:30–11:45	**A Lightweight Edge Intelligence Framework for Multi-Camera Vehicle Counting and Vehicle Analytics** Pankaj Kamboj · PK · **P0042**	**X-Band Compact High Gain Multichannel Transmitter and Down Converter with Integrated Monitoring and Reporting System** Yachamaneni Sandeep · PDIC · **P0018**	**Design and Simulation of Free-Flooded Ring Transducers for Sonar System** Basanta Roul · PDIC · **P00076**	**CSI-Based WiFi Sensing For Motion Detection Using Lightweight Signal Pre-processing and Unsupervised Learning** Nair Parvathy Arun · CRL-BG · **P0255**
11:45–12:00	**Training-Free Agentic Temporal Search for Long-Video Question Answering** Shubham Varshney · CRL-GAD · **P0062**	**Design Aspects of a Harmonic Tuned V/UHF Power Amplifier for Communication Radio** S Indumathi · CRL-BG · **P0188**	**Complex Signal Data Processing in RFPS** Dubbaka Shanker · HYD EW NS · **P0093**	**A Scalable Data Lakehouse Architecture for Ship Data Trend Analysis and Maritime Operational Intelligence** Srajan Ratti · CRL-GAD · **P0074**
12:00–12:15	**Operationally-Aware Explainability for Transformer-Based Automatic Target Recognition in UAV Imagery** Ritika Sinha · CRL-GAD · **P0075**	**Compact Ka-Band Monopulse Comparator with Co-Aligned Ports for Track Antenna** Sayyad Shahwaz Ali · CRL-BG · **P0327**	**AI- Based Predictive Maintenance for HUMSA Sonar Transducer Arrays Integrating Periodic Acoustic Characterization and Real-Time Telemetry** Kotresh Kumar DS · NS1 BG · **0079**	**Optimized Multi-Domain Feature Engineering for Seismic Activity Classification** Abhishek Kumar · CRL-GAD · **P0201**
12:15–12:30	**Category-Dependent Modality Reliance in Audio-Visual Violence Detection A Survey and a Proposed Efficient Fusion Framework** Gopesh kirad · CRL-BG · **P0183**	**An X-Band Two-Stage E-mode pHEMT Driver Amplifier for CORECHIP Integration** Mary Jennifer R · PDIC · **P0231**	**A Low Complexity Dual Stage Carrier Frequency Offset Implementation on FPGA - A Coarse-to-Fine Approach for Digital Radio Receivers** Shyamsundar Jha · CRL-BG · **P0181**	**Accuracy and Robustness in RF-Based Drone Identification Three Stress Tests on DroneRF** Priya Rawat · CRL-GAD · **P0291**
12:30–12:45	**An Explainable Hybrid Neural and Rule-Based Framework for Ship Category and Class Identification in ISAR Imagery** Sumit Srivastava · NCS-GAD · **P0209**	**Design and Validation of Parabolic Grid Reflector with Integrated Horn Feed** Amit Kumar Gupta · ANTENNA GAD · **P0090**	**Keystone Transform based method to resolve Doppler ambiguity in FMCW radar for multi-target scenario** Krishna Arvind Joshi · CRL-BG · **P0262**	**LLM-Guided World Modeling for Explainable Spatial Spectral Feature Classification** Hawa Singh · CRL-BG · **P0292**
12:45–13:00	**A Proposed Near-Infrared Face Recognition Framework for Low-Light Access Control Applications** Mohamed Facil T R · CRL-BG · **P0184**	**A Multi-Parametric Feature Engineering Approach for Predictive Health Monitoring of Dual-Motor Antenna Systems** Gaurav Kumar · ANTENNA GAD · **P0098**	**Signal Expansion using Burg Method for Improving FMCW Radar Range Resolution** Sherry Gomez · CRL-BG · **P0297**	**Real-Time RF Based Drone Detection and Angle of Arrival Estimation Using FPGA** Manish Kumar Pandey · CRL-GAD · **P0333**
13:00–13:15	**TransFusionDefender- An Adaptive Multi-Model Vision Transformer for Real-Time DDoS Mitigation in Edge Networks** Suraj Mishra · CRL-GAD · **P0050**	**Design of 60W Asymmetrical Doherty Power Amplifier for 5G applications** Kushan Chakraborty · CRL-BG · **P0355**	**Design and Experimental Validation of a CPFSK Modulator for Balise-BTM Communication** Alok Kumar · PDIC · **P0339**	**Offline Edge-AI-Based 2D/3D Motion Tracking and Activity Classification Using Inertial and Barometric Sensor** Naveen Kumar M · CRL-BG · **P0351**

DAY 1 · Wednesday, 11 November 2026 Technical Sessions 5–8 · 15:00–17:15 · Four Parallel Halls

TIME	HALL A · Dr. APJ Abdul Kalam Hall **Session 5: Machine Learning Applications** Chair: Shri. Rakesh Kumar, GM-BSTC Co-chair: Dr. Ramesh Chaveli, AGM-TP	HALL B · Sir C. V. Raman Hall **Session 6: RF Subsystems** Chair: Cmde. Srinivas (Retd), GM-A&A Co-chair: Shri. Rajkumar A, AGM PDIC	HALL C · Prof. Satish Dhawan Hall **Session 7: Product Design and Simulations** Chair: Shri. PT Kumaran, GM Comps Co-chair: Shri. Anil Kumar Patra AGM MR	HALL D · Sir M Visvesvaraya Auditorium **Session 8: Electronic Warfare** Chair: Shri. Srinivas K, ED-PDIC Co-chair: Smt. Charu S. Tripathi, AGM, CoE-EW&P
15:00–15:15	**SURAKSHA AI A Threat Prediction Model for the Early-Stage Software Development Lifecycle** Varun Srikakulam · SOFTWARE BG · **P0127**	**Design Of Fast Startup Wide Input Power Supply Unit For Railway Tele-Powering Applications** Venugopal B · PDIC · **P0110**	**Enhancing Thermal Uniformity in Radar Cold Plates through Passive Flow Redistribution** Dillip Mishra · MR BG · **P0039**	**Optimizing Emitter Direction Finding in Multi Panel BLI systems through PDW rejection** Naveen Kumar K · HYD EW NS · **P0084**
15:15–15:30	**Deep 3D U-Net Architecture for Hyperspectral Image Segmentation in Advanced Driver Assistance Systems** Vivek Kumar Singh Tomar · CRL-GAD · **P0169**	**An automatic Quiescent point stabilization for GaN devices** Anand M Tripathi · CRL-BG · **P0282**	**Design Optimization and Experimental Validation of an Underwater Propeller** Anurag Arjunan · PDIC · **P0143**	**Design and parametric optimization of composite reflector for antenna subjected to wind load** Krishn Kumar Gupta · ANTENNA GAD · **P0267**
15:30–16:00	**TEA BREAK & POSTER PRESENTATION VISIT**			
16:00–16:15	**Radar Emitter Classification Using Vector Embedded AI ML Models** Anurag · HYD EW NS · **P0096**	**Design and Analysis of Gain Equalizer For Wideband Cascaded RF Amplifiers** Dr Joydeb Mandal · CRL-BG · **P0260**	**Coupled Aerodynamics, Structural and Modal Analysis of 3D-Printed Wing Profile for Expendable Fixed-Wing UAVs** Nagappa · PDIC · **P00144**	**A Low Harmonic, Phase-Coherent x8 Frequency Clock Multiplier** Yenda Dilleswararao · PDIC · **P0198**
16:15–16:30	**MeteoCore - A State Driven Framework for Coordinated Multi-Effect Weather Rendering in Unreal Engine** Adarsh Kumar · CRL-GAD · **P0128**	**Highly Integrated X-Band Octal Transmit-Receive Module for Active Array Applications** Varsha G H · PDIC · **P0322**	**Numerical Estimation of Wind Drag and Dynamic Torque on a Rotating Shaped Parabolic Reflector** Vinay Sharma · ANTENNA GAD · **P0234**	**Dynamic Residual-Based Maximum-Likelihood Estimator for Bearings-Only Emitter Geolocation** Sheik Naseema · HYD EW NS · **P0100**
16:30–16:45	**A Comparative Study of Machine Learning Models for Predictive WiFi Roaming in High-Mobility Vehicular Environments** Anantha Bhargav · CRL-BG · **P0175**	**Compact Stripline Fed 8-way Radial Power Combiner for Ku Band SSPA** Kotla Rupasamyuktha · CRL-BG · **P0285**	**Supersonic Steady-State CFD Analysis of a Surface-to-Air Missile** Aadarsh Baish · CRL-BG · **P0328**	**Efficient Hardware Implementation of Cubic Spline Interpolation-Based Channel Estimation and TF Detection in ZP-OTFS Systems** Reenamol Jees · COE-COMM · **P0273**
16:45–17:00	**A Conceptual Multi-Scale CNN Transformer Architecture with Adaptive Feature Recalibration for Low Latency Automatic Modulation Classification in Electronic Warfare** Sada Vriksha · HYD EW LS · **P0223**	**Ultra-selective Bandpass Filter with High Stop Band Attenuation for LYNX U2 Fire Control System** Sayyad Shahwaz Ali · CRL-BG · **P0343**	**Finite Element Investigation of Rotary Frame Structural Response Under Varying Wind Loading Profiles** Arpit Garg · ANTENNA GAD · **P0342**	**Design and Development of Antenna Calibration Platform for Near-Field Test Range** Bangari Prudhviraj · ANTENNA GAD · **P0207**
17:00–17:15	**Mission-Aware Explainable Zero-Trust Authorization using Hybrid AI for Defense Systems** Karan Sonkar · SOFTWARE BG · **P0302**	**Multilayer Microstrip RHCP Antenna for IRNSS Applications** Sanu Roy · CRL-BG · **P0203**	**Pulsed Nd YAG Laser Hermetic Sealing of Aluminium Electronic Module Enclosures Design Rules and Process Implementation** Nehal Singh · COMP BG · **P0353**	**A Continuous Tunable True Time Delay Line Using Signal Interference Techniques** Dr Joydeb Mandal · CRL-BG · **P0256**

DAY 2 · Thursday, 12 November 2026 Technical Sessions 9–12 · 10:30–12:45 · Four Parallel Halls

TIME	HALL A · Dr. APJ Abdul Kalam Hall **Session 9: AI Powered Systems** Chair: Smt. Ruchi Pant, GM-TP Co-chair: Shri. Shashikant Yashwant Chaudhari, AGM-PDIC	HALL B · Sir C. V. Raman Hall **Session 10: Crypto & Cyber Security** Chair: Smt. Rekha Sastry, CTO-COMM Co-chair: Smt. Rekha Ashok Baradol, AGM,NWCS	HALL C · Prof. Satish Dhawan Hall **Session 11: Communication Systems** Chair: Shri. M Ramesh, CTO-EW&P Co-chair: Shri. Aman Choudhary, AGM, D&E-MCE	HALL D · Sir M Visvesvaraya Auditorium **Session 12: Radar Systems** Chair: Smt. Manjula Devi N, GM-MS Co-chair: Shri. Vidyanand P, AGM, D&E-MR
10:00-10:30	**TEA BREAK & POSTER PRESENTATION VISIT**			
10:30–10:45	**LOGIC Logistics Oriented Ground Intelligent Companion** Shrajan Jain · CRL-GAD · **P0027**	**A Kubernetes-Native Autonomous Cyber Defense Platform with MLOps and Hybrid Detection for On-Premises C5I Networks** Nishkarsh Patel · CRL-GAD · **P0045**	**Explanation-Stable Intrusion Detection, An XAI-Guided Adversarial Defense for Defense Network Security** Thejaswini Y R · SOFTWARE BG · **P0047**	**Range Migration Correction for Enhanced Detection of Hypersonic Targets Using Shear Transform** Sriharsha Nag T S · CRL-BG · **P0043**
10:45–11:00	**An Intelligent Enterprise Lakehouse Framework for Metadata-Aware Data Engineering and AI-Driven Analytics** Anuj Kumar · CRL-GAD · **P0086**	**A Multi-Context Aware SOAR Framework Using Retrieval-Augmented Generation for Intelligent Incident Response** Rajat · CRL-GAD · **P0102**	**User-centric SIEM Approach for Enhanced Insider Threat Analysis and Detection** Anmol Raj · CRL-GAD · **P0070**	**Design and Implementation of a Modular Python-Based Ethernet Control and Monitoring System for High-Power Amplifiers with Integrated Calibration and Diagnostics** Abhay Kumar Yadav · NS2 BG · **P0044**
11:00–11:15	**AI Data Shield - An Intelligence Driven Approach to Preventing Data Leakage** Mayank Jatav · CRL-GAD · **P0132**	**Agentic Cognitive Electronic Warfare - A Multi-Agent Reinforcement Learning-MARL Framework for Autonomous Radar Deception and Spectrum Dominance** Mukesh Prajapat · CRL-GAD · **P0073**	**A secure encryption middleware for file transfer on private network of IPFS and Hyperledger fabric** K Anu Ayyappan · CRL-BG · **P0149**	**GPGPU based Real-time Signal Processing for AESA Radars** K. Revati Suresh · CRL-BG · **P0140**
11:15–11:30	**Explainable AI for Predictive Maintenance in Industrial Automation** Yaman Kurrey · PDIC · **P0134**	**A Formally Specified Parallel and Adaptive Architecture for Diehard Randomness Testing** Anurag Parashar · CRL-BG · **P0218**	**Decentralized QoS Forecasting in Resource Constrained Ad Hoc Networks using Federated MLP** Amnajasmin K · CRL-BG · **P0221**	**Practical Methodology for Generating Site-Specific Monopulse Calibration Tables from Operational SSR Reply Data** Ankur Singh · RADAR GAD · **P0193**
11:30–11:45	**Consistency over Relevance A Controlled Study of Source-Consistency Hallucination in Multi-Source Retrieval-Augmented Generation** Chirde Vaibhav Rajesh · CRL-BG · **P0168**	**A Generic Runtime and Station Specific Plugin Architecture for Configurable Computer Based Interlocking Systems** Ansh Kuma · CRL-GAD · **P0233**	**PAPR Reduction Techniques for Code Multiplexed DSSS based Data Link Systems** Yerrolla Pavan Kumar · CRL-BG · **P0250**	**Programmable Waveform and Timing Generator Module for 3D Surveillance Radar** Tejal Mali · NS-2 BG · **P0226**
11:45–12:00	**Reward-Guided Lightweight Neural Architecture Search using Double Deep Reinforcement Learning** Vyayavaram Venkat Sai Rohith · PDIC · **P0228**	**A Hybrid Encryption Model for Secure Military IoT Communication Using RSA/ECC and AES-256-GCM** Neeraj Solanki · HYD EW NS · **P0243**	**Survey on Anti Jamming & Anti Spoofing Algorithms for GNSS Quad Band CRPA Antenna** Surya Narayanan S · CRL-BG · **P0178**	**Design of a Scalable Heterogeneous Computing System for Real-Time Perimeter Surveillance Radar** Kurella Durga Rao · CRL-BG · **P0242**
12:00–12:15	**Intelligent Skies AI Driven Controller Pilot Speech Recognition for Next-Generation Air Traffic Management** Chandramani Adil · CRL-GAD · **P0232**	**WAFShield A Machine Learning Based Web Application Firewall for Accurate HTTP Attack Detection** Dharmendra Tanwar · CRL-GAD · **P0275**	**Performance Analysis of 5G NR PDSCH over NTN Channels with Integrated Doppler Pre-compensation** S R Deekshith Prasad · CRL-BG · **P0277**	**A Reconfigurable FPGA-Based Digital Beamforming Architecture Using Real-Time Parameter Streaming for Scalable Phased Array Systems** Nirbhay Kumar Singh · COE-R&WS · **P0249**
12:15–12:30	**Knowledge-Enhanced MCQA, Dynamic Knowledge Graph Integration, Option Reasoning, and Confidence Scoring** Arunima Roy · CRL-BG · **P0321**	**Realization of Quantum Random number generator by using Arrival Time of the photon for high security applications** Rishabh Bahoria · CRL-BG · **P0305**	**Raspberry Pi Controlled PIN-Diode Reconfigurable Intelligent Surface for Wireless Communications** Subham Biswas · CRL-BG · **P0280**	**Range Migration Correction for High-Speed Maneuvering Multi Targets via Second-Order Keystone Transform and Generalized De-Chirp Processing** Karthikeyan P · COE- R&WS · **P0265**
12:30–12:45	**Physics-Guided Neural State Estimation** Yerram Deekshith Kumar · CRL-GAD · **P0301**	**A Layered PKI Framework for Automated Certificate Lifecycle Management Across Heterogeneous Endpoints in Large Enterprise Networks** Surabhi Ashok Babu · CRL-BG · **P0016**	**FPGA Realization of Real-Time PCMA Systems for Satellite Communication Links** Divya N · CRL-BG · **P0308**	**Design & Implementation of Reconfigurable P4 Polyphase Based Radar Waveform Generator** Raghavendra Singh · RADAR GAD · **P0269**

DAY 2 · Thursday, 12 November 2026 Technical Sessions 13–16 · 15:15–17:30 · Four Parallel Halls

TIME	HALL A · Dr. APJ Abdul Kalam Hall **Session 13: Cognitive Systems** Chair: Shri. Rajendra Sambrani, I/C GM, Seekers/RF&IR Co-chair: Shri. Paresh Chandra Nayak, AGM, Seeker	HALL B · Sir C. V. Raman Hall **Session 14: Smart Computing Systems** Chair: Shri. Jagadish Kumar G, GM-CHN Co-chair: Shri Venkateswara Rao G, AGM EW&A	HALL C · Prof. Satish Dhawan Hall **Session 15: Autonomous Systems** Chair: Shri. Hemant Kumar Gupta, GM-US Co-chair: Shri. Rajesh Motta N, AGM-CHN	HALL D · Sir M Visvesvaraya Auditorium **Session 16: Photonics Systems** Chair: Shri. Dhirendra N Pandey, GM-MC Co-chair: Shri. Ravindra Y. Chaudhari, AGM, D&E-LS/PN
15:15–15:30	**Physics-Informed AI for Time-Based Separation - Optimizing Runway Throughput via Wake Vortex Dynamics** Nikhil Shukla · CRL-GAD · **P0299**	**Numerical Simulation of Aerosol Transport and Collection Efficiency in an Air Contamination Detection System Using CFD techniques** Vivek Thite · BELOP · **P0020**	**HSTDG AM MAPPO A Cognitive Swarm Intelligence Framework for Autonomous Search and Strike Operations** Vinod Kumar · CRL-GAD · **P0024**	**Optical design of shared aperture SWIR/VNIR optical system for Long Range Spotter application** Arjun · COE-EO&P · **P0155**
15:30–15:45	**HFCR-6G Hybrid Federated Cognitive Radio Framework for Intelligent Anti-Jamming and Spectrum Management in 6G Networks** Avik Das · MILCOM BG · **P0247**	**Machine Learning-Based Prediction of PCB Natural Frequency Using Geometric Dimensions and Copper Percentage** Lokeswara Rao Morla · PDIC · **P0021**	**Interaction-Time Feasibility and Risk Estimation for Safe Teleoperated Robotic Manipulation in Unknown Environment** Pooja Goswami · CRL-GAD · **P0026**	**Photonic-Assisted RF Self-Interference Cancellation System for 5G In-Band Full-Duplex Transceivers** Divya Sree K · CRL-BG · **P0165**
15:45–16:00	**Area Efficient M Term Karatsuba Decomposition for Finite Field Multiplier** Tamilarasan A · CRL-BG · **P0274**	**CPU-Optimized S-57 ENC Visualization with Topology-Preserving LOD and Radar PPI Fusion** Gautam Verma · CRL-GAD · **P0082**	**Policy-Materialized Access Control in a Multi-Tenant C4I Maritime Domain Awareness** Akash Chahar · CRL-GAD · **P0041**	**A Quantum Noise Based True Random Number Generator Utilizing Stochastic Detector Fluctuations** Naresh Sothuku · CRL-BG · **P0200**
16:00–16:15	**Enhancing Thermal Dissipation Paths in Hermetically Sealed Underwater Electronics for Improved System Reliability** Karuppaiah R · PDIC · **P0354**	**A Multi-protocol Fail-Safe Bootloader Framework for Remote Firmware Updates in Automatic Train Protection Systems** Abhinav Pathak · CRL-GAD · **P0103**	**A Modular and Numerically Robust Statistical Gating Framework for Real-Time Radar Multi-Target Tracking** Rakshith G M · COE-R&WS · **P0118**	**Comparative Analysis of Enhancement and Thresholding Methods for Single Frame Small Infrared Target Detection** Hima C · CRL-BG · **P0230**
16:15–16:30	**Performance Evaluation of GENEX Guidance for Impact-Angle-Constrained Trajectory Shaping in Glide Bombs** Bhagya S Prasanth · CRL-BG · **P0350**	**High-Capacity Multi-Format Media Server with Dynamic Pipeline Construction, Automated NAT Traversal, and UAV Geospatial Fusion** Anushka Uniyal · CRL-GAD · **P0109**	**Real-Time LiDAR-Based Perception for Static and Dynamic Obstacle Detection and Tracking on GNSS-Only Autonomous Surface Vessels** Fahad A M · CRL-BG · **P0158**	**Automated Detection of Small Airborne Targets in Electro-Optical Infrared Imagery Using Multi Frame Input** Anubhav Mittal · NCS BG · **P0257**
16:30–16:45	**Adaptive Receiver Noise Estimation for Bayesian Direction Finding Using Spinning and Omnidirectional Antennas** Rohith Joshua Gumpula · HYD EW LS · **P0288**	**Flame Photometry based Real-Time Detection and Classification of Biological Threats in an Environment** Killa Kishore Babu · CRL-BG · **P0148**	**Evolution of Air Surveillance Radar Processing Architectures An Architectural Perspective on Convergence with Multifunction Radar Systems** Avinash Kaur · RADAR GAD · **P0159**	**Photonics-based Multi-Band Coherent Transceiver for Radar Systems** Rupam Biswas · CRL-BG · **P0264**
16:45–17:00	—	**An Integrated Framework for Automated CIS Benchmark Compliance on Red Hat Enterprise Linux** Pradeep Kumar Tiwari · CRL-BG · **P0150**	**A Generic Modular Software Architecture for the Report Processing Layer in Modern Radar Systems** Shailendra Kumar · CRL-GAD · **P0161**	**Indigenous Development of Ultra-High Vacuum Dewar Assembly for Cooled Thermal Imaging Systems** Vivek Thite · BELOP · **P0284**
17:00–17:15	—	**Hard Real-time Timers in Linux on x86 Architecture** Sreenivas S · CRL-BG · **P0182**	**Dual-Mode Path-Following Control of an Underactuated Autonomous Underwater Vehicle** Kommuri Hima Bindu · CRL-BG · **P0272**	**Comparison of Virtual Image formation through prism and bulk optics architecture and its analysis in Head Level Display** Abhishek Tiwari · PDIC · **P0317**
17:15–17:30	—	**Heterogeneous Computing Architecture for AI-Enabled ELINT Processors** Dileep Kumar · HYD EW NS · **P0095**	**Visual Model-Based Automation of Traction SCADA HMI and DSS** Sneha Prajapati · CRL-GAD · **P0336**	**Design, Fabrication and MIL-STD Qualification of a Fiber Optic Uplink/Downlink Network for Digital MFSTAR Radar Featuring Sub-30 ns Optical Skew** Chetan Aivalli · ADSN BG · **P0326**

Session chairs and co-chairs All sessions at a glance

Session	Topic	Hall	Day · Time	Chair	Co-chair
1	Computer Vision	Hall A	Day 1 · 11:30–13:15	Smt. Deepika Gupta, GM, DCCS-GAD	Shri. Nandakumar S, AGM-NS
2	Microwave Components	Hall B	Day 1 · 11:30–13:15	Shri. Vrushabhendra S H M, In-GM, ANTENNA-GAD	Shri. Prakash S P, AGM-PDIC
3	Signal Processing	Hall C	Day 1 · 11:30–13:15	Shri. Mini Nair S, GM-NS1	Shri. Jeyagopal P AGM Mkt MR
4	Detection & Classification	Hall D	Day 1 · 11:30–13:15	Smt. Seema Yadav, CS, CRL-GAD	Shri. M Sadasivaiah AGM EW&A
5	Machine Learning Applications	Hall A	Day 1 · 15:00–17:15	Shri. Rakesh Kumar, GM-BSTC	Dr. Ramesh Chaveli, AGM-TP
6	RF Subsystems	Hall B	Day 1 · 15:00–17:15	Cmde. Srinivas (Retd), GM-A&A	Shri. Rajkumar A, AGM PDIC
7	Product Design and Simulations	Hall C	Day 1 · 15:00–17:15	Shri. PT Kumaran, GM Comps	Shri. Anil Kumar Patra AGM MR
8	Electronic Warfare	Hall D	Day 1 · 15:00–17:15	Shri. Srinivas K, ED-PDIC	Smt. Charu S. Tripathi, AGM, CoE-EW&P
9	AI Powered Systems	Hall A	Day 2 · 10:30–12:45	Smt. Ruchi Pant, GM-TP	Shri. Shashikant Yashwant Chaudhari, AGM-PDIC
10	Crypto & Cyber Security	Hall B	Day 2 · 10:30–12:45	Smt. Rekha Sastry, CTO-COMM	Smt. Rekha Ashok Baradol, AGM, NWCS
11	Communication Systems	Hall C	Day 2 · 10:30–12:45	Shri. M Ramesh, CTO-EW&P	Shri. Aman Choudhary, AGM, D&E-MCE
12	Radar Systems	Hall D	Day 2 · 10:30–12:45	Smt. Manjula Devi N, GM-MS	Shri. Vidyanand P, AGM, D&E-MR
13	Cognitive Systems	Hall A	Day 2 · 15:15–17:30	Shri. Rajendra Sambrani, In-GM, Seekers/RF&IR	Shri. Paresh Chandra Nayak, AGM, Seeker
14	Smart Computing Systems	Hall B	Day 2 · 15:15–17:30	Shri. Jagadish Kumar G, GM-CHN	Shri Venkateswara Rao G, AGM EW&A
15	Autonomous Systems	Hall C	Day 2 · 15:15–17:30	Shri. Hemant Kumar Gupta, GM-US	Shri. Rajesh Motta N, AGM-CHN
16	Photonics Systems	Hall D	Day 2 · 15:15–17:30	Shri. Dhirendra N Pandey, GM-MC	Shri. Ravindra Y. Chaudhari, AGM, D&E-LS/PN